

VLSI Programming – Comprehensive Syllabus

Course Overview

This course provides a comprehensive introduction to Very Large Scale Integration (VLSI) design, covering digital IC design, CMOS technology, Hardware Description Languages (HDL), verification, synthesis, physical design, Design-for-Test (DFT), and ASIC design flow. Students will gain practical experience using industry-standard EDA tools and develop VLSI projects from specification to chip implementation.

Duration: 16 Weeks (80–100 Hours)

Prerequisites:

- Basic Digital Electronics
 - Boolean Algebra
 - Basic Electronics
 - Fundamentals of Computer Architecture (recommended)
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Module 1: Introduction to VLSI Technology (Week 1)

Topics

- Introduction to VLSI
- History and Evolution
- VLSI Design Flow
- ASIC vs FPGA
- Moore's Law
- IC Fabrication Process
- Semiconductor Basics
- CMOS Technology Overview

Practical Exercises

- EDA Tool Installation
- Introduction to VLSI Design Environment
- Basic Digital Circuit Analysis

Module 2: CMOS Fundamentals (Week 2)

Topics

- MOSFET Operation
- CMOS Inverter
- Logic Gates using CMOS
- Static & Dynamic Power
- Propagation Delay
- Noise Margins
- Fan-In & Fan-Out

Practical Exercises

- CMOS Gate Design
 - Power and Delay Analysis
 - Logic Optimization
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Module 3: Verilog HDL Programming (Week 3)

Topics

- Introduction to Verilog
- Module Structure
- Data Types
- Operators
- Procedural Blocks
- Conditional Statements
- Loops
- Functions
- Tasks

Practical Exercises

- Basic Verilog Programs
 - Combinational Logic Design
 - Arithmetic Logic Unit (ALU)
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Module 4: Sequential Logic Design (Week 4)

Topics

- Flip-Flops
- Registers
- Counters
- Shift Registers
- Finite State Machines (FSM)
- Mealy & Moore Models

Practical Exercises

- Counter Design
 - Sequence Detector
 - Traffic Light Controller
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Module 5: Simulation & Functional Verification (Week 5)

Topics

- Testbenches
- Simulation Techniques
- Waveform Analysis
- Functional Verification
- Debugging HDL Code

Practical Exercises

- ALU Verification
 - FSM Simulation
 - Counter Testing
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Module 6: Logic Synthesis (Week 6)

Topics

- RTL Design
- Gate-Level Design
- Logic Optimization
- Timing Constraints
- Synthesis Reports
- Resource Utilization

Practical Exercises

- RTL to Gate-Level Conversion
 - Synthesis Report Analysis
 - Timing Verification
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Module 7: Static Timing Analysis (STA) (Week 7)

Topics

- Timing Concepts
- Setup Time
- Hold Time
- Clock Skew
- Clock Jitter
- Timing Violations
- Timing Closure

Practical Exercises

- Timing Report Analysis
 - Setup/Hold Violation Fixes
 - Clock Constraint Implementation
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Module 8: Physical Design Fundamentals (Week 8)

Topics

- Floorplanning
- Power Planning
- Placement
- Clock Tree Synthesis (CTS)
- Routing
- Design Rule Check (DRC)
- Layout vs Schematic (LVS)

Practical Exercises

- Floorplan Creation
 - Placement Optimization
 - Routing Analysis
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Module 9: Design for Testability (DFT) (Week 9)

Topics

- Scan Chains
- Boundary Scan
- JTAG
- Built-In Self-Test (BIST)
- Fault Models
- Automatic Test Pattern Generation (ATPG)

Practical Exercises

- Scan Chain Design
 - Fault Simulation
 - Test Pattern Generation
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Module 10: Low Power VLSI Design (Week 10)

Topics

- Power Optimization Techniques

- Clock Gating
- Multi-Voltage Design
- Power Gating
- Dynamic Voltage Scaling (DVS)
- Leakage Reduction

Practical Exercises

- Clock Gating Implementation
 - Power Analysis
 - Low-Power RTL Design
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Module 11: ASIC Design Flow (Week 11)

Topics

- Complete ASIC Design Flow
- RTL Design
- Verification
- Synthesis
- Physical Design
- Sign-Off
- Tape-Out Process

Practical Exercises

- End-to-End ASIC Flow
 - Sign-Off Checks
 - Design Optimization
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Module 12: Verification Methodologies (Week 12)

Topics

- Introduction to Verification
- Assertion-Based Verification
- Universal Verification Methodology (UVM) Basics
- Functional Coverage
- Code Coverage

- Regression Testing

Practical Exercises

- Assertion Writing
 - Functional Coverage Analysis
 - Verification Testbench Development
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Module 13: Mini Project (Week 13)

Choose one:

- Digital Calculator
 - Traffic Light Controller
 - UART Controller
 - Elevator Controller
 - ALU Design
 - FIFO Memory
 - SPI Controller
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Module 14–16: Capstone Project

Complete VLSI Design Project

- Requirement Analysis
 - RTL Design
 - Functional Verification
 - Logic Synthesis
 - Static Timing Analysis
 - Physical Design
 - DRC & LVS Verification
 - Power Optimization
 - Documentation
 - Final Presentation
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Tools & Software

- Verilog HDL
 - SystemVerilog (Introduction)
 - Cadence Virtuoso
 - Cadence Genus
 - Cadence Innovus
 - Synopsys Design Compiler
 - Synopsys PrimeTime
 - Mentor Graphics ModelSim / QuestaSim
 - Xilinx Vivado (for FPGA Prototyping)
 - Intel Quartus Prime (optional)
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Core Concepts Covered

- Digital IC Design
 - CMOS Circuit Design
 - RTL Coding
 - Functional Verification
 - Static Timing Analysis (STA)
 - Logic Synthesis
 - Physical Design
 - Low-Power Design
 - Design for Testability (DFT)
 - ASIC Design Flow
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Assessment Scheme

Component	Weightage
Assignments	15%
Lab Exercises	25%
Quizzes	10%
Mini Project	20%
Capstone Project	30%

Learning Outcomes

By the end of the course, learners will be able to:

- Understand CMOS technology and VLSI design principles.
- Develop digital hardware designs using Verilog HDL.
- Design and verify combinational and sequential logic circuits.
- Perform RTL simulation, synthesis, and timing analysis.
- Apply physical design concepts including floorplanning, placement, routing, DRC, and LVS.
- Implement Design-for-Test (DFT) techniques and optimize designs for power and performance.
- Understand the complete ASIC design flow from specification to tape-out.
- Use industry-standard Electronic Design Automation (EDA) tools for VLSI development.
- Build industry-ready digital IC designs suitable for semiconductor, automotive, telecommunications, consumer electronics, aerospace, and high-performance computing applications.